Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

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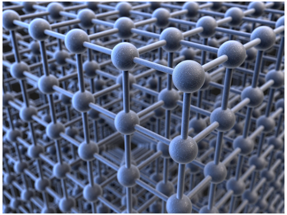
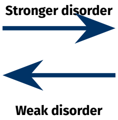
Non-linear phenomena

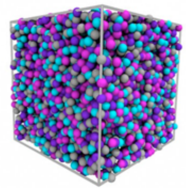
Crystal Transitions and Disorder

Crystalline





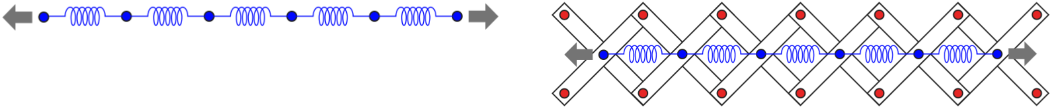
Amorphous



Meta-materials

spring-chain

spring-chain system embedded into a pantograph



Scientific Context: Nonlinearity & Instability

Scientific Challenges

The Challenge:

Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

- **Phase Transitions:** Martensitic transformations.
- **Plasticity:** Dislocation-mediated flow and avalanches.
- **Fracture:** Brittle-to-ductile transitions and damage.

Unifying Perspective

These phenomena are inherently **displacement-driven** (*non-diffusive*) and can be studied within the **same context** of displacive instabilities.